INCH-POUND

MIL-DTL-13074C
15 January 2009
SUPERSEDING
MIL-DTL-13074B
27 June 2003

DETAIL SPECIFICATION

WIRE, ELECTRICAL, ANTENNA, (WIRE W-27 AND WIRE WS-19( ) /U)

Inactive for new design after 16 June 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers stranded, round electrical wire fabricated from silicon bronze, suitable for antenna applications.

1.2 Classification. Silicon bronze wires covered by this specification are of the following types (see 6.2).

1.2.1 Wire W-27. Wire W-27 consists of 7 strands of 0.064 inch diameter silicon bronze wire approximately equivalent to 26.300 circular mils (6 AWG) in area.

1.2.2 Wire WS-19( )/U. Wire WS-19( )/U consists of 7 strands of 0.040 inch diameter silicon bronze wire approximately equivalent to 10,400 circular mils (10 AWG) in area (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Columbus, ATTN: DSCC-VAI, P.O. Box 3990, Columbus, Ohio 43218-3990, or email to WireCable@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 6145

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the in the solicitation or contract (see 6.2).

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 17025	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of these documents are available online at http://www.iso.ch or from the International Organization for Standardization American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

NATIONAL CONFERENCE OF STANDARDS LABORATORIES (NCSL)

NCSL Z540.3	Requirements for the Calibration of Measuring and Test Equipment

(Copies of this document is available online at http://www.ncsli.org or from the National Conference of Standards Laboratories (NCSL), 2995 Wilderness Place, Suite 107, Boulder, CO 80301-5404.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2 Materials. The material for the wire shall be as specified herein.

3.3 Strands.

3.3.1 Wire W-27. Each strand of wire W-27 shall consist of a silicon bronze alloy wire with a diameter of 0.064 inch (1.63 mm), ± 0.0015 inch (.04 mm) (see 3.4.1).

3.3.2 Wire WS-19( )/U. Each strand of wire WS-19( )/U shall consist of a silicon bronze alloy wire with a diameter of 0.040 inch (1.02 mm), ± 0.0015 inch (.04 mm) (see 3.4.2).

3.4 Conductor construction.

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3.4.1 Wire W-27. Wire W-27 shall be constructed of 7 strands of wire as specified in 3.3.1, twisted with a left-hand lay of a maximum 3 inches in length.

3.4.2 WS-19( )/U. Wire WS-19( )/U shall be constructed of 7 strands of wire as specified in 3.3.2 twisted with a left-hand lay of a maximum 2 inch in length.

3.5 Conductor joints. Joints in conductors should be avoided, and in no case shall they exceed two joints per strand per length of wire, with joints spaced not less than 3 feet apart. All joints shall be butt brazed using silver alloy solder and a nonacid flux; and shall be finished smoothly in a workmanlike manner. No twist joints shall be used. The tensile strength of a section of each strand which includes a joint, shall not be less than 80 percent of the tensile strength of an adjacent section of the strand of equal length without a joint. The average tensile strength of a group of twelve strands which include joints, shall not be less than 90 percent of the adjacent sections of the strands (see 4.6).

3.6 Breaking strength (see 4.6).

3.6.1 Wire W-27. Wire W-27 shall have a breaking strength of not less than 2200 pounds.

3.6.2 Wire WS-19( )/U. Wire WS-19( )/U shall have a breaking strength of not less than 790 pounds.

3.7 Elongation. The elongation shall be not less than 0.80 percent for a 60 inch length of stranded conductor.

3.8 Workmanship. Strands and wire shall be manufactured and finished in a careful and workmanlike manner in accordance with the good design and sound practice.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Conformance inspection (see 4.4)

b. Qualification inspection (see 4.3)

4.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspections shall be used. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with NCSL Z540.3 and ISO 17025 or equivalent.

4.3 Qualification inspection. Each operator required to make joints in strands will submit a group of twelve specimens of strand which include a joint with a specimen of adjacent normal section of each. These specimens will be tested for breaking strength. An operator will be considered qualified after submitting a group of twelve specimens which meet the requirements of 3.5. Records of such qualifications shall be available for verification by the Government.

4.4 Conformance inspection. Conformance inspection shall consist of groups A and B inspections (see 4.4.5 and 4.4.6, respectively) and shall be performed on every lot of cable procured under this specification. Sampling inspection shall be accomplished for each lot in accordance with 4.4.2.

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4.4.1 Lot. A lot shall consist of all cable manufactured under substantially the same conditions and offered for inspection at one time.

4.4.1.1 Lot size. The lot size shall be defined as the number of units of product submitted for inspection.

4.4.1.2 Unit of product. A unit of product shall be defined as the continuous length of cable contained on a single reel, spool, or in a package.

4.4.1.3 Specimen. A specimen is a single piece of finished wire which is taken from a sample unit and subjected to inspection.

4.4.2 Sampling. A random sample shall be selected from each lot specified in table I.

TABLE I. Inspection sample.

Inspection lot size 1/	Sample size
1	1
2 to 8	2
9 to 90	3
91 to 150	12
151 to 280	19
281 to 500	21
501 to 1,200	27
1,201 to 3,200	36
3,201 to 10,000	38
10,001 to 35,000	46

1/ Lot size is based on the number of units of product.

4.4.3 Rejected lot. Failure of any sample to pass any inspection shall constitute a failure of the lot. If an inspection lot is rejected, the contractor may rework the lot to correct the defects or screen out the defective units, and resubmit the lot for re-inspection. Such lots shall be separated from new lots and shall be identified as re-inspected lots (see 4.4.4).

4.4.4 Noncompliance. If a sample fails to pass any inspection, the contractor shall notify the cognizant inspection activity of such failure and take corrective action on the materials or processes or both, as warranted on all units of the product. Acceptance and shipment of the product shall be discontinued until corrective action has been taken. After the corrective action has been taken, the conformance inspection shall be repeated on replacement articles. (This includes all tests and examinations, or only the test that the original sample failed, at the option of the cognizant inspection activity.) Final acceptance and shipment shall be withheld until inspection has shown that the corrective action was successful. In the event of failure after re-inspection, information concerning the failure shall be provided to the cognizant inspection activity.

4.4.5 Group A inspection. This inspection, including sampling, shall consist of visual dimensional examinations of paragraph 4.5 to determine compliance with the requirements of 3.2 through 3.5 and 3.8 as applicable.

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4.4.6 Group B inspection. This inspection, including sampling, shall consist of the tests and examinations of 4.6.

4.5 Visual and dimensional inspection. Strands and completed wire shall be inspected to verify that the materials and all workmanship comply with this specification.

4.5.1 Construction and lay. Construction and lay shall be determined by physical examination and measurement of the complete wire.

4.6 Breaking strength and elongation. Breaking strength and elongation shall be made on a tensile strength machine, preferably of the pendulum types. Breaking strength test shall be made after stranding, on the completed wire. The free length between grips of the test specimen shall be not less than 60 inches (1524.00 mm) and the speed shall be not more than 3 inches (76.20 mm) per minute. Care shall be taken to insure that strands in the stranded wire are evenly gripped during test. (Successfully testing of stranded wire for tensile properties requires an adequate means of gripping the ends of the test specimen. Various means are available, such as a long tube or socket into which the wire may be soldered, or in which, after insertion, the wires may be swaged or pressed without serious distortion. Ordinary jaws or clamping devices are usually not suitable). At the conclusion of the test, the wire shall meet the requirements for breaking strength and elongation specified in 3.5 and 3.6, respectively.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Wire covered by this specification is most commonly used in antenna applications.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. If required, the specific issue of individual documents referenced (see 2.1).

c. Packaging requirements (see 5.1).

d. Classification of wire required (see 1.1).

e. Length of wire required.

f. Size of spool or reel and length on each (see 5.1).

h. Coil, spool, and reel marking requirements.

6.3 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: WS-19W/U. As soon as possible after the award of contract, the contractor should apply to the Government office specified in the contract for such information.

6.4 Subject term (key word) listing.

Round
Silicon
Bronze
Stranded

6.5 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmental Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website http://www.epa.gov/osw/hazard/wastemin/priority.htm. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (see section 3).

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2008-144)

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